

EXERCISE (O-1)

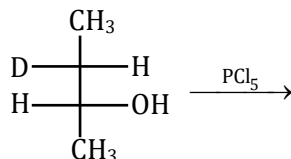
SINGLE CORRECT TYPE QUESTIONS

Preparation of alkyl halides from alcohol and PCl_3 , PCl_5 , SOCl_2 & SOCl_2 with pyridine

1. The number of moles of PCl_5 required to prepare Ethyl chloride from ethyl alcohol is:
 (A) 2 (B) 1 (C) 4 (D) 3

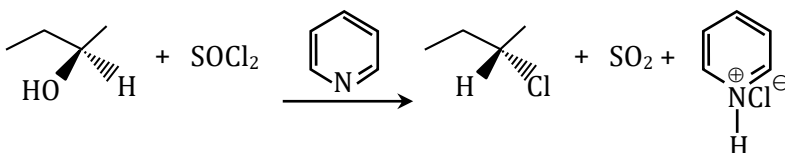
CHL080

2. Total number of stereoisomers of the major product formed in the following reaction is:



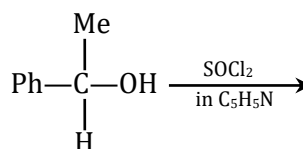
- (A) 3 (B) 4 (C) 5 (D) 6

CHL081

3. The reaction  proceeds by the mechanism

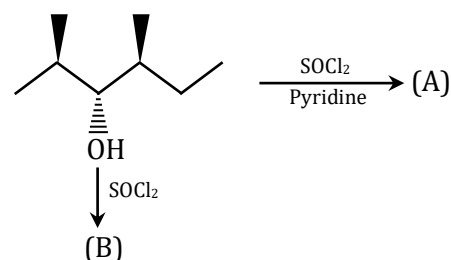
- (A) $\text{S}_{\text{N}}1$ (B) $\text{S}_{\text{N}}2$ (C) $\text{S}_{\text{N}}\text{i}$ (D) E2

CHL082

4.  Which statement is true for the reaction ?

- (A) Retention of configuration (B) Inversion of configuration
 (C) Inversion and Retention both (D) None

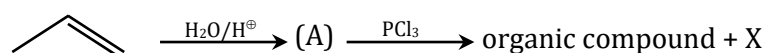
CHL083

5. 

The relationship in between (A) and (B) is :

- (A) Enantiomers (B) Identical (C) Diastereomers (D) Geometrical isomers

CHL084

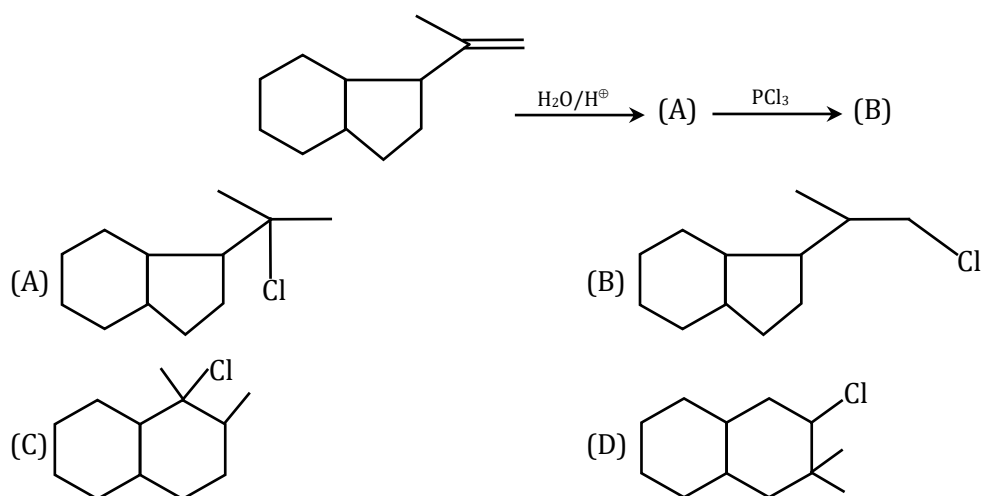
6. 

Which statement is correct for X.

- (A) X is monobasic acid (B) X has oxidation state +1
 (C) X has oxidation state +3 (D) X is tribasic acid

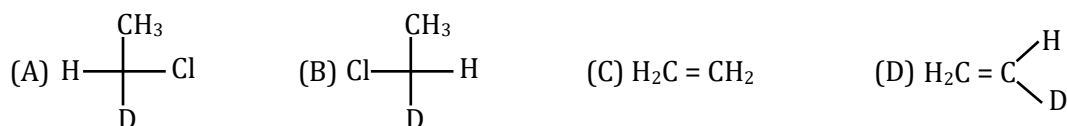
CHL085

7. In the following reaction find product B.



CHL086

8. $\text{H}-\overset{\text{CH}_3}{\underset{\text{D}}{\text{C}}}-\text{OH} \xrightarrow[\text{Pyridine}]{\text{SOCl}_2}$ (A). The product A will be :



CHL087

9. The reaction of SOCl_2 on alkanols to form alkyl chlorides gives good yields because
- (A) Alkyl chlorides are immiscible with SOCl_2
- (B) The other products of the reaction are gaseous and escape out
- (C) Alcohol and SOCl_2 are soluble in water
- (D) The reaction does not occurs via intermediate formation of an alkyl chloro sulphite.

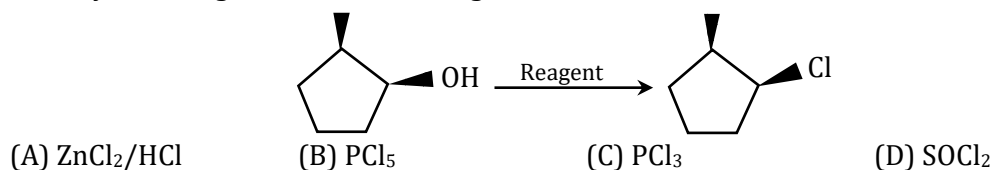
CHL088

10. $\text{H}-\overset{\text{CH}_3}{\underset{\text{D}}{\text{C}}}-\text{OH} \xrightarrow[\text{Et}_2\text{O}]{\text{SOCl}_2}$ (A). The product A will be :



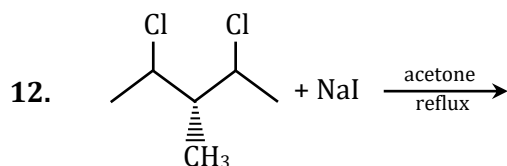
CHL089

11. Identify best reagent for the following conversion



CHL090

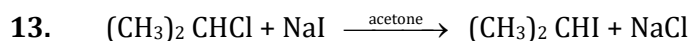
Finkelstein reaction & Swarts reaction



The total number of products (including stereo isomerism) in the above reaction are :

- (A) 2 (B) 6 (C) 4 (D) 8

CHL092



The above reaction is known as :

- (A) Perkin reaction (B) Fitting reaction
(C) Finkelstein reaction (D) Sabatier and Sanderson's reaction

CHL093

14. Which of the following cannot be used for Swarts reaction ?

- (A) AgF (B) Hg_2F_2 (C) CoF_2 (D) HSbF_6

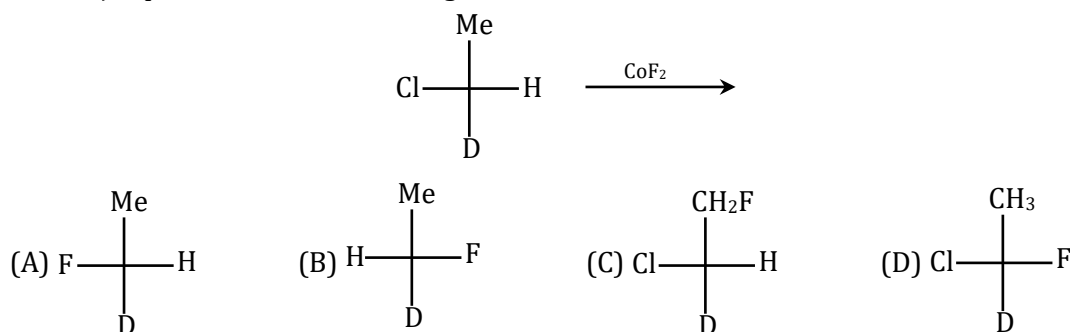
CHL094

15. The driving force for Swart's Reaction is :

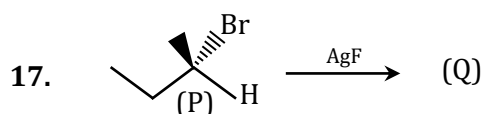
- (A) Higher electronegativity of F atom (B) Higher solubility of metal fluorides
(C) Higher bond dissociation energy of C-F (D) Lower solubility of metal fluorides

CHL095

16. The Major product for the following reaction is :



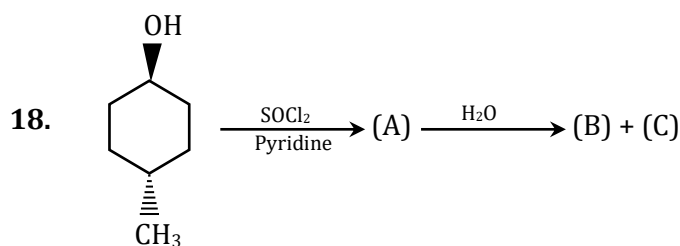
CHL096



The configuration of P and Q are :

- (A) R, S (B) S, R (C) R, R (D) S, S

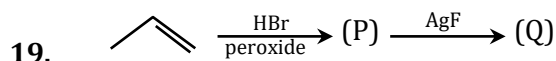
CHL097



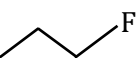
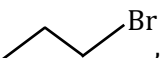
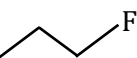
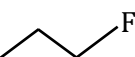
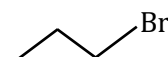
The relationship in between (B) and (C) is :

- (A) Enantiomers (B) Optical isomers (C) Diastereomers (D) Functional isomers

CHL098



P and Q are respectively :

- (A) ,  (B) , 
- (C) ,  (D) , 

CHL099

20. Which of the following is NOT a reason for using dry acetone in Finkelstein reaction?

- (A) NaCl or NaBr formed is precipitated in dry acetone.
(B) It shifts the equilibrium in forward direction
(C) Solubility of NaI is high in dry acetone
(D) It helps in dissociation of C-Cl or C-Br bond.

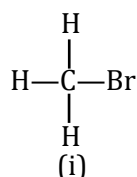
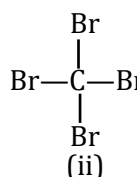
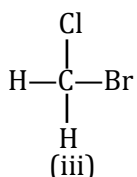
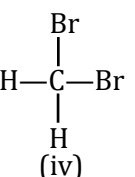
CHL100

21. Which of the following has density lower than water?

- (A) Bromo alkanes (B) Poly Chloro alkanes
(C) Iodo alkanes (D) Fluoro alkanes

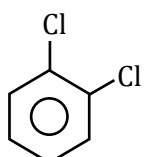
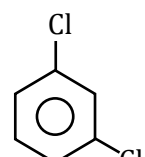
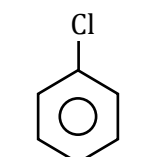
CHL101

22. The correct order of increasing boiling points for the following compounds

-  (i)  (ii)  (iii)  (iv)
- (A) II < I < III < IV (B) I < IV < II < III
(C) I < III < IV < II (D) IV < III < II < I

CHL102

23. The correct order of melting point -

-  (A)  (B)  (C)
- (A) A > B > C (B) B < A < C (C) B < A < C (D) A < C < B

CHL103

24. Which of the following has the highest boiling point

- (A) CH₃CH₂I (B) CH₃Cl (C) CH₃I (D) CH₃Br

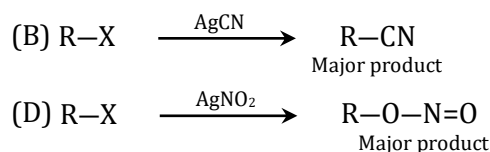
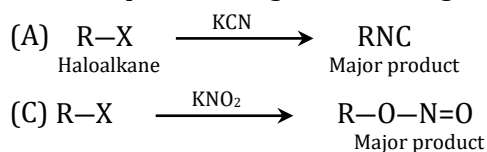
CHL104

25. Which of the following has highest dipole moment:

- (A) CH₃Cl (B) CH₃F (C) CH₃Br (D) CH₃I

CHL105

26. Correct option among the following :



CHL107

27. Iodoform is used as

- (A) anaesthetic (B) antiseptic (C) analgesic (D) anti fibrin

CHL108

28. The following is used in paint removing

- (A) CHCl_3 (B) CH_2Cl_2 (C) CCl_4 (D) CH_3Cl

CHL109

29. In fire extinguishers, following is used

- (A) CHCl_3 (B) CS_2 (C) CCl_4 (D) CH_2Cl_2

CHL110

30. The following is used for metal cleaning and finishing

- (A) CHCl_3 (B) CHI_3 (C) CH_2Cl_2 (D) C_6H_6

CHL111

31. First chlorinated insecticide

- (A) DDT (B) Gammaxene (C) Iodoform (D) Freon

CHL112

32. The following is used as anaesthetic

- (A) C_2H_4 (B) CHCl_3 (C) CH_2Cl_2 (D) DDT

CHL113

33. Freon-12 is

- (A) CF_3Cl (B) CHCl_2F (C) CF_2Cl_2 (D) CFCl_3

CHL114

34. The name of DDT

- (A) p,p-dichloro diphenyl trichloro ethane (B) p,p-dichloro diphenyl trichloro ethene
 (C) p,p-dichloro diphenyl trichloro benzene (D) p,p-tetra chloro ethane

CHL115

35. Freon R-22 is

- (A) CHClF_2 (B) CCl_2F_2 (C) CH_3Cl (D) CH_2Cl_2

CHL116

36. Molecular formula of DDT has

- (A) 5 Cl atoms (B) 4 Cl atoms (C) 3 Cl atoms (D) 2 Cl atoms

CHL117

37. What is DDT among the following

- (A) Greenhouse gas (B) A fertilizer
 (C) Biodegradable pollutant (D) Non-Biodegradable pollutant

CHL118

38. The compound used in the production of Freon refrigerant, R-22 is

- (A) methylene chloride (B) chloroform
 (C) iodoform (D) carbon tetra chloride

CHL119

39. The compound used as an anaesthetic during surgery is
 (A) chloroquine (B) halothane (C) chloramphenicol (D) thyroXIIne

CHL120

40. Assertion (A) : n-Butyl chloride has higher boiling point than tert-butyl chloride.
 Reason (R) : C-Cl bond in n-butyl chloride is more polar than in tert-butyl chloride.
 (A) Both A and R are true and R is the correct explanation to A
 (B) Both A and R are true and R is not the correct explanation to A
 (C) A is true but R is false
 (D) A is false but R is true

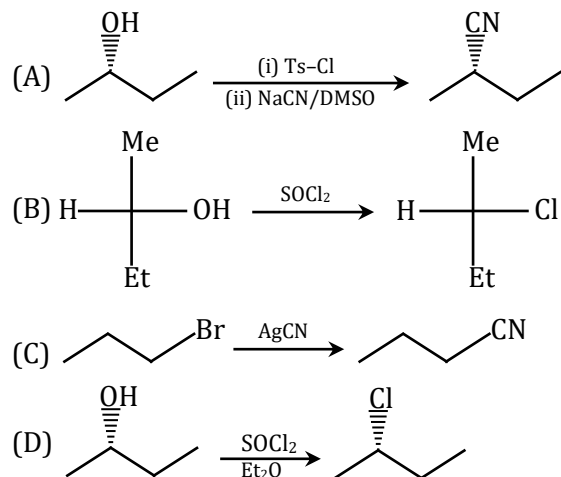
CHL121

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Preparation of alkyl halides from alcohol and PCl_3 , PCl_5 , SOCl_2 & SOCl_2 with pyridine

1. Identify reaction incorrectly matched with their major product -



CHL091

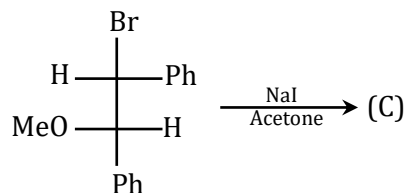
EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

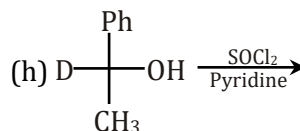
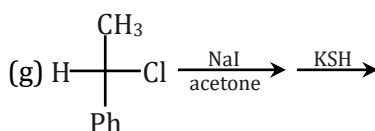
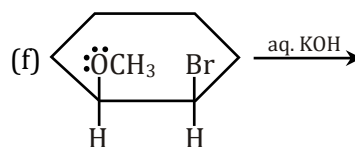
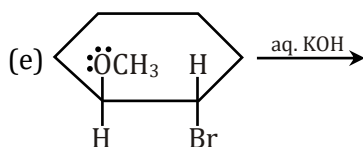
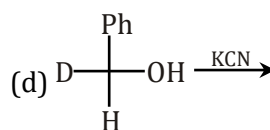
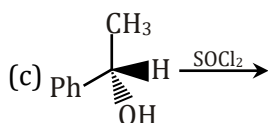
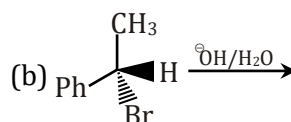
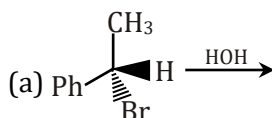
Finkelstein reaction

1. Number of chiral centre present in product (C) in the following reaction sequence :



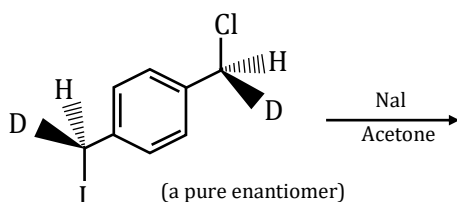
CHL106

2. Find the number of reaction those proceed with retention of configuration.



CHL123

3. In the following reaction, how many different S_N2 products would be formed?



CHL124

4. If 2, 4-dichloropentane is treated with one equivalent of C_2H_5ONa in ethanol, only one HCl is eliminated to give chloro pentene. How many different chloro pentanes (including stereo-isomers) would be formed.

CHL125

5. If 3-chlorobutanol is treated with $\text{C}_2\text{H}_5\text{ONa}/\text{C}_2\text{H}_5\text{OD}$, it undergo E_1CB elimination. How many different butenal would be formed?

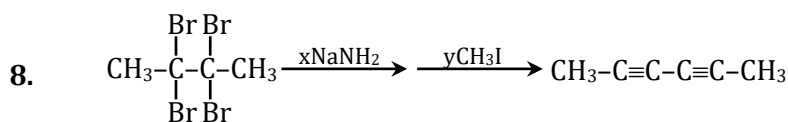
CHL126

6. 1,3-dichlorocyclopentane exist in three stereo isomeric forms, of which only two are chiral. If one mole of its achiral stereoisomer is treated with exactly one mole of NaOH in aqueous medium and only mono substitution ($\text{S}_\text{N}2$) and mono-elimination($\text{E}2$) occur producing chloro-cyclopentanol and chlorocyclopentene, how many different products are expected ?

CHL127

7. If 1-chloro-4-methylcyclohexane is treated with KOH (aq), both substitution ($\text{S}_\text{N}2$) and elimination ($\text{E}2$) reaction occur simultaneously in comparable amount. In total, how many substitution plus elimination products would result?

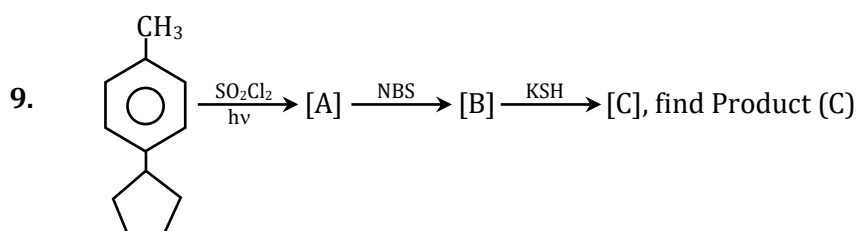
CHL128



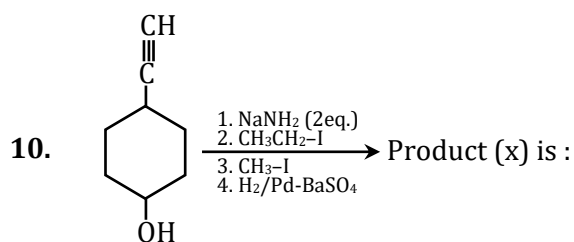
x and y mole consumed.

Value of $x+y=?$

CHL129



CHL130



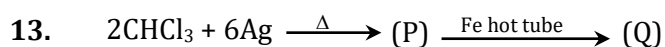
CHL131

11. The number of structurally isomeric tribromo derivatives possible for benzene

CHL026

12. How many structural isomers of $\text{C}_5\text{H}_{11}\text{Br}$ are primary alkyl halide?

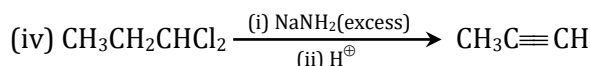
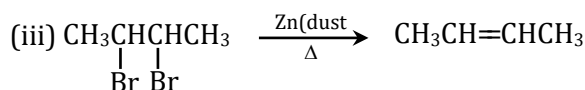
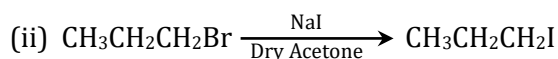
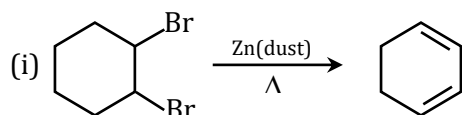
CHL027



Degree of unsaturation of Q is :

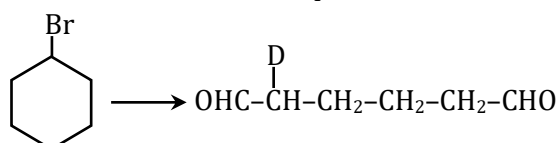
CHL028

14. Number of reaction in Which product is wrong (major) product.



CHL029

15. Minimum number of steps involved in conversion of



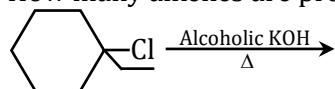
CHL196

Part-2 : JEE Advanced (Decimal type)

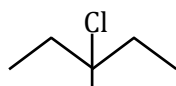
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Elimination Reaction and Substitution Reaction

16. How many alkenes are produced in the following elimination reaction



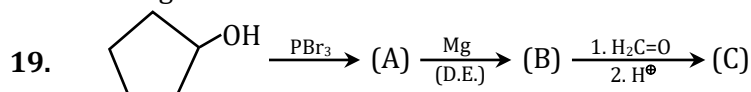
CHL189

17. Total numbers of possible alkene obtained on reaction of  with alcoholic KOH.

CHL190

18. The total number of alkenes possible by dehydrobromination of 3-bromo-3-cyclopentylhexane using alcoholic KOH is

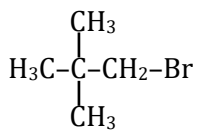
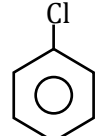
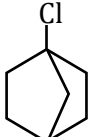
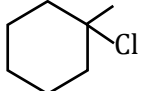
CHL191



What is the molecular weight of C.

CHL192

20. Out of following how many compounds are undergoes to $\text{S}_{\text{N}}2$ reaction.

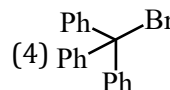
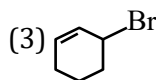
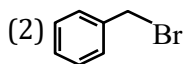
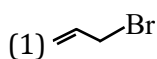
(i)		(ii)	$\text{CH}_2=\text{CH}-\text{Cl}$	(iii)	
(iv)	$\text{Ph}-\text{CH}_2-\text{Cl}$	(v)		(vi)	

CHL193

EXERCISE - JEE (Main) PYQ

1. Which among the following halides will generate the most stable carbocation in Nucleophilic substitution reaction?

[JEE (Main) 2025]



CHL201

2. Given below are two statements :

Statement I : $\text{CH}_3 - \text{O} - \text{CH}_2 - \text{Cl}$ will undergo $\text{S}_{\text{N}}1$ reaction though it is a primary halide.

Statement II : $\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} - \text{CH}_2 - \text{Cl}$ will not undergo $\text{S}_{\text{N}}2$ reaction very easily though it is a primary halide.

In the light of the above statements, choose the **most appropriate answer** from the options given below:

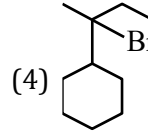
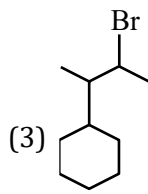
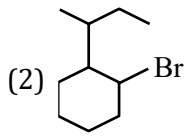
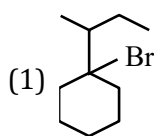
[JEE (Main) 2025]

- (1) **Statement I** is incorrect but **Statement II** is correct.
 (2) Both **Statement I** and **Statement II** are incorrect
 (3) **Statement I** is correct but **Statement II** is incorrect
 (4) Both **Statement I** and **Statement II** are correct.

CHL202

3. When sec-butylcyclohexane reacts with bromine in the presence of sunlight, the major product is:

[JEE (Main) 2025]



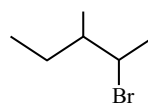
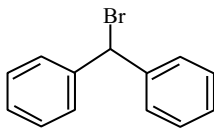
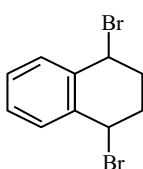
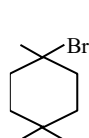
CHL203

4. The compound with molecular formula C_6H_6 , which gives only one monobromo derivative and takes up four moles of hydrogen per mole for complete hydrogenation has _____ π electrons.

[JEE (Main) 2025]

CHL204

5. The ascending order of relative rate of solvolysis of following compounds is



(A)

(B)

(C)

(D)

[JEE (Main) 2025]

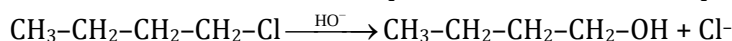
- (1) $(\text{D}) < (\text{A}) < (\text{B}) < (\text{C})$
 (3) $(\text{D}) < (\text{B}) < (\text{A}) < (\text{C})$

- (2) $(\text{C}) < (\text{B}) < (\text{A}) < (\text{D})$
 (4) $(\text{C}) < (\text{D}) < (\text{B}) < (\text{A})$

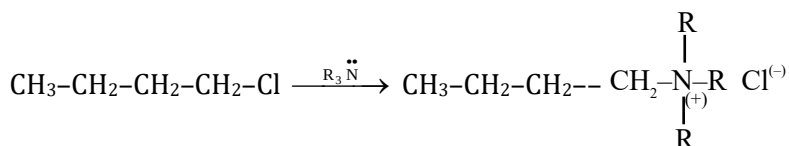
CHL205

6. Given below are two statements :

Statement-I : The conversion proceeds well in the less polar medium.



Statement-II : The conversion proceeds well in the more polar medium.



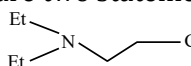
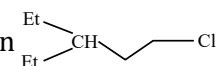
In the light of the above statements, choose the *correct* answer from the options given below.

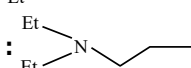
[JEE (Main) 2025]

- (1) Both statement I and statement II are true
- (2) Both statement I and statement II are false
- (3) Statement I is false but statement II is true
- (4) Statement I is true but statement II is false

CHL206

7. Given below are two statements :

Statement I :  will undergo alkaline hydrolysis at a faster rate than 

Statement II :  Cl, intramolecular substitution takes place first by involving lone pair of electrons on nitrogen.

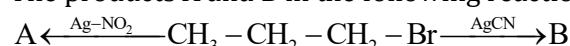
In the light of the above statements, choose the **most appropriate** answer from the options given below :

[JEE (Main) 2025]

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both Statement I and Statement II are correct
- (4) Statement I is correct but Statement II is incorrect

CHL207

8. The products A and B in the following reactions, respectively are

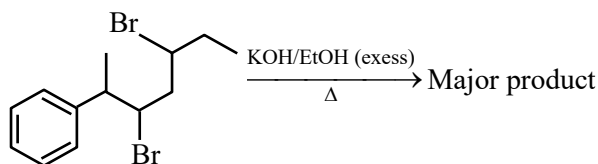


[JEE (Main) 2025]

- (1) $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--ONO}$, $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--NC}$
- (2) $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--ONO}$, $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--CN}$
- (3) $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--NO}_2$, $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--CN}$
- (4) $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--NO}_2$, $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--NC}$

CHL208

9. The major product of the following reaction is :

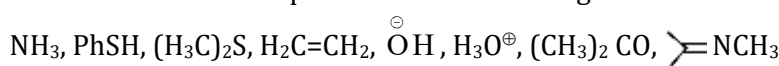


[JEE (Main) 2025]

- (1) 6-Phenylhepta-2,4-diene
- (2) 2-Phenylhepta-2,5-diene
- (3) 6-Phenylhepta-3,5-diene
- (4) 2-Phenylhepta-2,4-diene

CHL209

10. Total number of nucleophiles from the following is :-

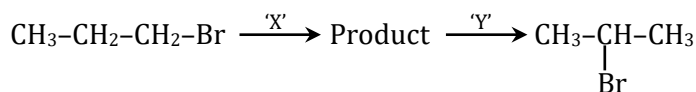


[JEE (Main) 2025]

- (1) 5
- (2) 4
- (3) 7
- (4) 6

CHL210

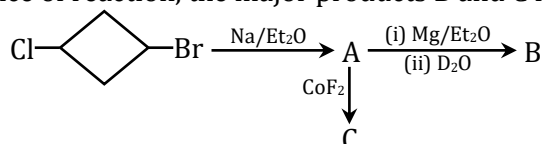
11. Identify the correct set of reagents or reaction conditions 'X' and 'Y' in the following set of transformation. [JEE (Main) 2024]



- (1) X = conc.alc. NaOH, 80°C, Y = Br₂/CHCl₃
 (2) X = dil.aq. NaOH, 20°C, Y = HBr/acetic acid
 (3) X = conc.alc. NaOH, 80°C, Y = HBr/acetic acid
 (4) X = dil.aq. NaOH, 20°C, Y = Br₂/CHCl₃

CHL194

12. In the following sequence of reaction, the major products B and C respectively are :



- (1) D----D and F----F
 (2) D----D and F----F
 (3) D----D and F----F
 (4)  and F----F

[JEE (Main) 2024]

CHL195

13. **Assertion A :** Hydrolysis of an alkyl chloride is a slow reaction but in the presence of NaI, the rate of the hydrolysis increases. [JEE (Main) 2023]

Reason R : I⁻ is a good nucleophile as well as a good leaving group.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is false but R is true
 (2) A is true but R is false
 (3) Both A and R are true and R is the correct explanation of A
 (4) Both A and R are true but R is NOT the correct explanation of A

CHL132

14. Identify the correct order for the given property for following compounds [JEE (Main) 2023]

(A) Boiling point : Cl < Cl < Cl

(B) Density : Br < Cl < I

(C) Boiling point : Br < Br < Br

(D) Density : I < Br < Cl

(E) Boiling point : Cl > Cl > Cl

Choose the correct answer from the option given below :-

- (1) (B), (C) and (D) only (2) (A), (C) and (E) only
 (3) (A), (C) and (D) only (4) (A), (B) and (E) only

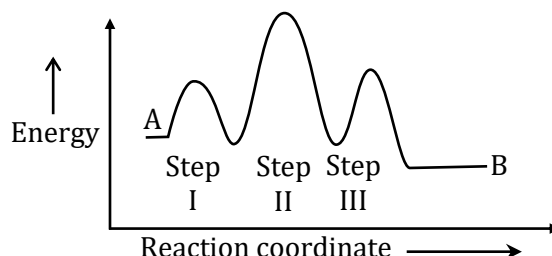
CHL135

15. In the following halogenated organic compounds the one with maximum number of chlorine atoms in its structure is : [JEE (Main) 2023]
- (1) Chloral (2) Gammexene
(3) Chloropicrin (4) Freon -12

CHL138

16. Consider the following reaction that goes from A to B in three steps as shown below:-

[JEE (Main) 2023]

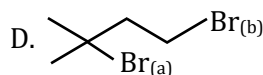
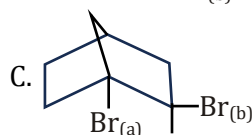
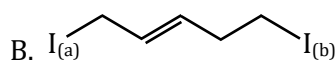
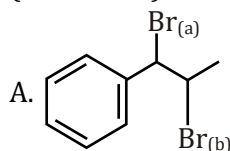


Choose the correct

	Number of intermediates	Number of Activated complexes	Rate determining step
(1)	3	2	II
(2)	2	3	II
(3)	2	3	I
(4)	2	3	III

CHL141

17. Choose the halogen which is most reactive towards S_N1 reaction in the given compounds (A, B, C & D) [JEE (Main) 2023]



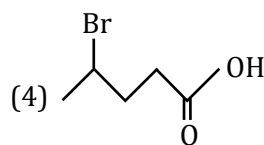
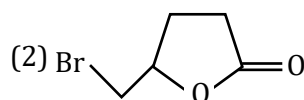
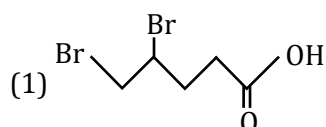
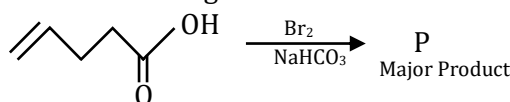
- (1) A-Br(b) ; B-I(b) ; C-Br(b) ; D-Br(b)
(3) A-Br(b) ; B-I(a) ; C-Br(a) ; D-Br(a)

- (2) A-Br(a) ; B-I(a) ; C-Br(b) ; D-Br(a)
(4) A-Br(b) ; B-I(a) ; C-Br(a) ; D-Br(a)

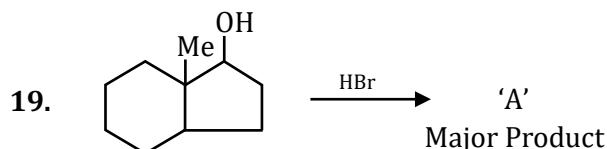
CHL142

18. Major product 'P' formed in the following reaction is :-

[JEE (Main) 2023]



CHL143

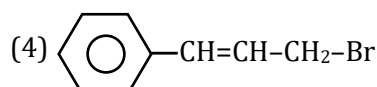
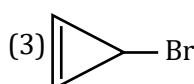
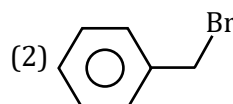
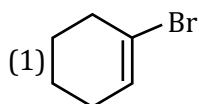


The number of hyperconjugation structures involved to stabilize carbocation formed in the above reaction is _____. [JEE (Main) 2023]

CHL145

20. Compound from the following that will not produce precipitate on reaction with AgNO_3 is :

[JEE (Main) 2023]



CHL147

21. 2-Methyl propyl bromide reacts with $\text{C}_2\text{H}_5\text{O}^-$ and gives 'A' whereas on reaction with $\text{C}_2\text{H}_5\text{OH}$ it gives

'B'. The mechanism followed in these reactions and the products 'A' and 'B' respectively are:

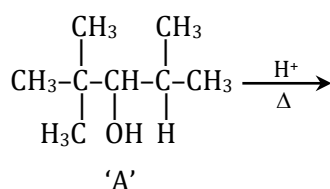
[JEE (Main) 2023]

- (1) $\text{S}_{\text{N}}2$, A = iso-butyl ethyl ether; $\text{S}_{\text{N}}1$, B = tert-butyl ethyl ether
 (2) $\text{S}_{\text{N}}1$, A = tert-butyl ethyl ether; $\text{S}_{\text{N}}1$, B = 2-butyl ethyl ether
 (3) $\text{S}_{\text{N}}1$, A = tert-butyl ethyl ether; $\text{S}_{\text{N}}2$, B = iso-butyl ethyl ether
 (4) $\text{S}_{\text{N}}2$, A = 2-butyl ethyl ether; $\text{S}_{\text{N}}2$, B = iso-butyl ethyl ether

CHL148

22. For the given reaction

[JEE (Main) 2023]



The total number of possible products formed by tertiary carbocation of A is _____.

CHL150

23. Match List I with List II

[JEE (Main) 2023]

1 - Bromopropane is reacted with reagents in List I to give product in List II

	LIST I - Reagent		LIST II - Product
A.	KOH (alc)	I.	Nitrile
B.	KCN (alc)	II.	Ester
C.	AgNO_2	III.	Alkene
D.	H_3CCOOAg	IV.	Nitroalkane

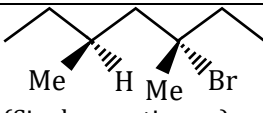
- (1) A-IV, B-III, C-II, D-I
 (3) A-I, B-II, C-III, D-IV

- (2) A-III, B-I, C-IV, D-II
 (4) A-I, B-III, C-IV, D-II

CHL151

EXERCISE - JEE (Advanced) PYQ

1. Match the reactions in List-I with the features of their products in List-II and choose the correct option. [JEE (Advanced) 2023]

List-I		List-II	
(P)	(-)-1-Bromo-2-ethylpentane $\xrightarrow[\text{S}_\text{N}2 \text{ reaction}]{\text{aq. NaOH}}$ (single enantiomer)	(1)	Inversion of configuration
(Q)	(-)-2-Bromopentane $\xrightarrow[\text{S}_\text{N}2 \text{ reaction}]{\text{aq. NaOH}}$ (single enantiomer)	(2)	Retention of configuration
(R)	(-)-3-Bromo-3-methylhexane $\xrightarrow[\text{S}_\text{N}1 \text{ reaction}]{\text{aq. NaOH}}$ (single enantiomer)	(3)	Mixture of enantiomers
(S)	 (Single enantiomer) $\xrightarrow[\text{S}_\text{N}1 \text{ reaction}]{\text{aq. NaOH}}$	(4)	Mixture of structural isomers
		(5)	Mixture of diastereomers

(A) P $\rightarrow$ 1; Q $\rightarrow$ 2; R $\rightarrow$ 5; S $\rightarrow$ 3

(B) P $\rightarrow$ 2; Q $\rightarrow$ 1; R $\rightarrow$ 3; S $\rightarrow$ 5

(C) P $\rightarrow$ 1; Q $\rightarrow$ 2; R $\rightarrow$ 5; S $\rightarrow$ 4

(D) P $\rightarrow$ 2; Q $\rightarrow$ 4; R $\rightarrow$ 3; S $\rightarrow$ 5

CHL154

2. The reaction of 4-methyloct-ene (**P**, 2.52 g) with HBr in the presence of $(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2$ gives two isomeric bromides in a 9 : 1 ratio, with combined yield of 50%. Of these, the entire amount of the primary alkyl bromide was reacted with an appropriate amount of diethylamine followed by treatment with eq. K_2CO_3 to give a non-ionic product **S** in 100% yield.

The mass (in mg) of **S** obtained is ____.

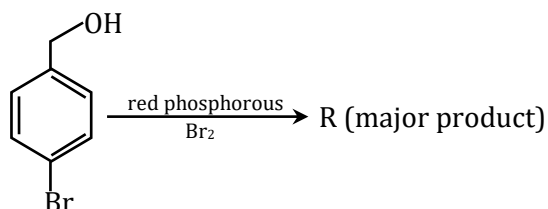
[Use molar mass (in g mol^{-1}) : H = 1, C = 12, N = 14, Br = 80]

[JEE (Advanced) 2023]

CHL155

3. Consider the following reaction.

[JEE (Advanced) 2022]



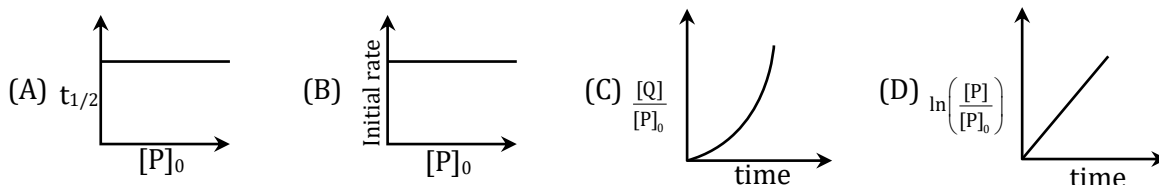
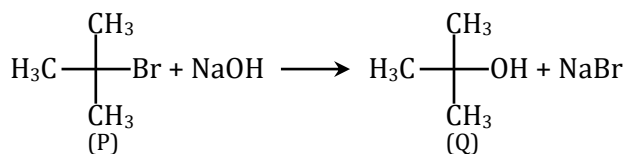
On estimation of bromine in 1.00 g of **R** using Carius method, the amount of AgBr formed (in g) is ____.

[Given : Atomic mass of H = 1, C = 12, O = 16, P = 31, Br = 80, Ag = 108]

CHL153

4. Which of the following plots is(are) correct for the given reaction?
 ($[P]_0$ is the initial concentration of P)

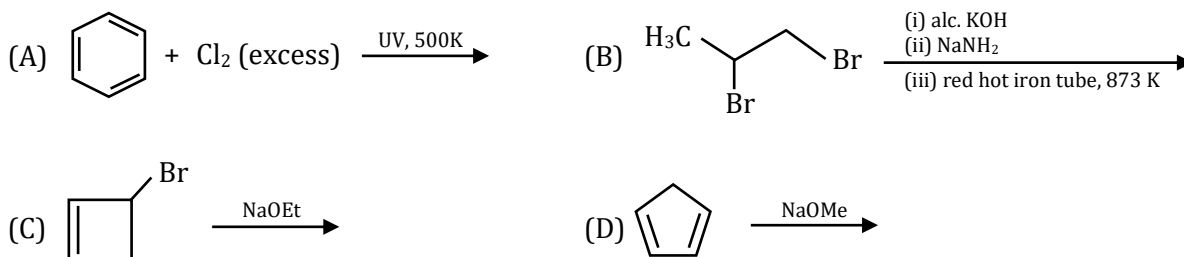
[JEE (Advanced) 2020]



CHL078

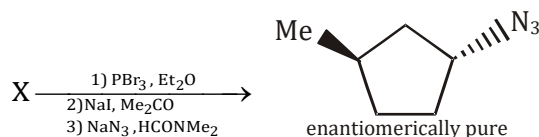
5. Choose the correct option(s) that give(s) an aromatic compound as the major product.

[JEE (Advanced) 2019]



CHL152

6. In the following reaction sequence, the correct structure(s) of X is (are):



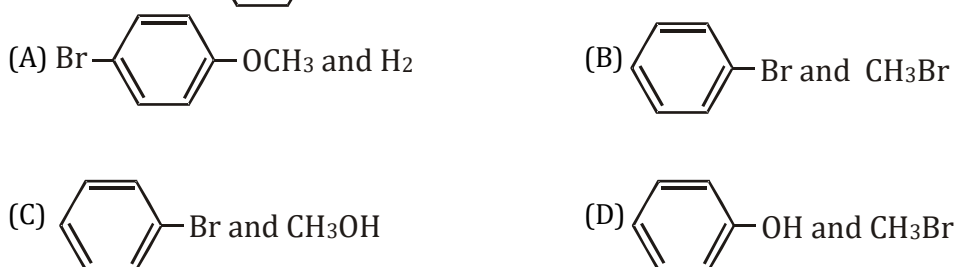
[JEE (Advanced) 2018]



CHL077

7. In the reaction $\xrightarrow{\text{HBr}}$ the products are

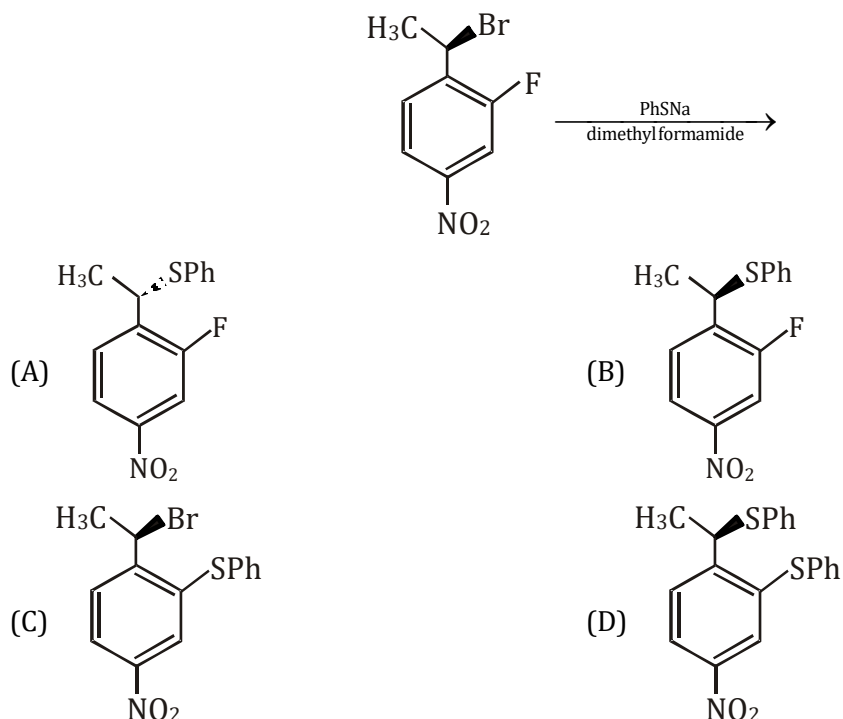
[IIT-2010]



CHL076

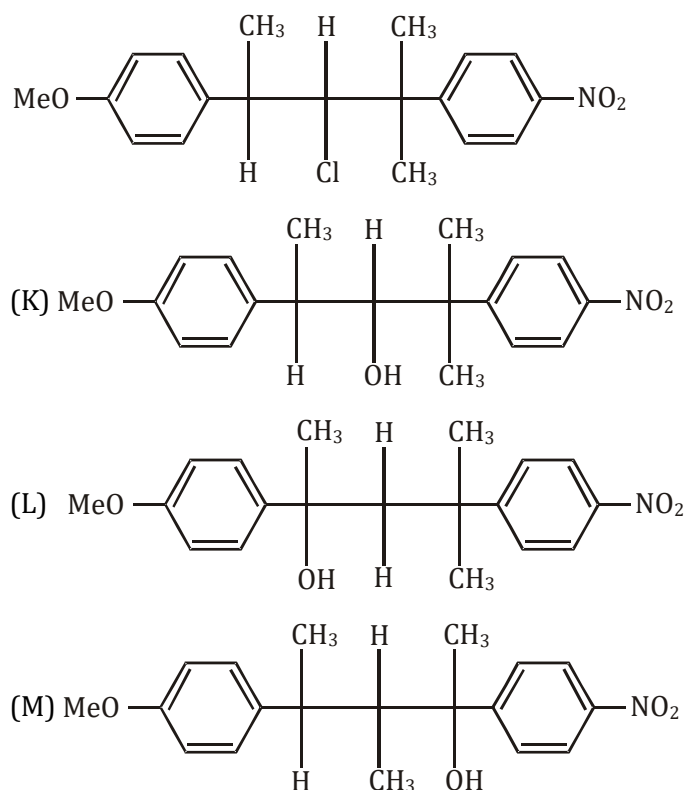
8. The major product of the following reaction is

[IIT -2008]



9. The following compound on hydrolysis in aqueous acetone will give:

CHL075
[IIT - 2005]



It mainly gives

(A) K and L

(B) Only K

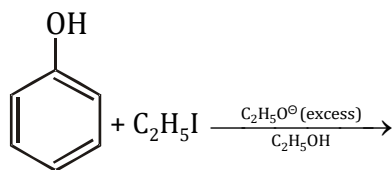
(C) L and M

(D) Only M

CHL074

10. The product of following reaction is :

[IIT -2003]



(A) $\text{C}_6\text{H}_5\text{OC}_2\text{H}_5$

(B) $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$

(C) $\text{C}_6\text{H}_5\text{OC}_6\text{H}_5$

(D) $\text{C}_6\text{H}_5\text{I}$

CHL073

11. Arrange the following compounds in order of increasing dipole moment

[IIT 1996]

Toluene m-dichlorobenzene o-dichlorobenzene p-dichlorobenzene

(I)

(II)

(III)

(IV)

(A) $\text{I} < \text{IV} < \text{II} < \text{III}$

(B) $\text{IV} < \text{I} < \text{II} < \text{III}$

(C) $\text{IV} < \text{I} < \text{III} < \text{II}$

(D) $\text{IV} < \text{II} < \text{I} < \text{III}$

CHL198

ANSWER KEY

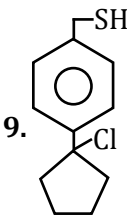
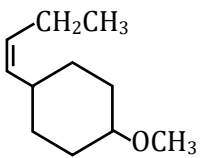
EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	B	B	C	C	C	B	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	C	C	D	B	B	B	C	B	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	C	B	A	A	C	B	B	C	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	B	C	A	A	A	D	B	B	C

EXERCISE (O-2)

Que.	1	
Ans.	A,C	

EXERCISE (S)

1. (2) 2. (4) 3. (4) 4. (6) 5. (4)
6. (5) 7. (4) 8. (8)
9. 
10. 
11. (3) 12. (4) 13. (4) 14. (1)
15. (4) 16. (2.00) 17. (3.00) 18. (5.00) 19. (100.00)
20. (1.00)

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	4	8	1	1	3	4	4	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	3	2	2	2	2	2	7	1
Que.	21	22	23							
Ans.	1	4	2							

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	1791	1.50	A	B,D	B	D	A	A	B
Que.	11									
Ans.	B									